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from nltk.tokenize import sent_tokenize, word_tokenize
from nltk import pos_tag
from nltk.tag import hmm
import nltk, random

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nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')

```

```

text = """
I love learning new things
Machine learning is interesting
Data science helps students
I enjoy learning models
Artificial intelligence is powerful
"""

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sentences = sent_tokenize(text)

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tagged_sentences = []
for sent in sentences:
    words = word_tokenize(sent)
    tagged_sentences.append(pos_tag(words))

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print("Tagged sentences:", tagged_sentences)

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trainer = hmm.HiddenMarkovModelTrainer()
hmm_model = trainer.train_supervised(tagged_sentences)

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print("States learned:", hmm_model._states)

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def generate_sentence(model, length=10):
    sentence = []
    state = random.choice(list(model._states))

    for _ in range(length):
        word = model._outputs[state].generate()
        sentence.append(word)
        state = model._transitions[state].generate()

    return sentence

```

```

generated = generate_sentence(hmm_model, 10)
print("Generated sentence:")
print(" ".join(generated))

```

```

Tagged sentences: [[('I', 'PRP'), ('love', 'VBP'), ('learning', 'VBG'), ('new', 'JJ'), ('things', 'NNS'), ('Machine', 'NNP')]
States learned: ['PRP', 'VBP', 'VBG', 'JJ', 'NNS', 'NNP', 'NN', 'VBZ']
Generated sentence:
new models Machine intelligence is new things Data intelligence helps
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!

```